RESOLUTION OF THE DIAGONAL SHEAF BY BOX PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. One-step reduction to the on-product card (same batch); the closed-immersion observation is in-span.

Allowed standard prerequisites (editorial).

- Closed immersion coherence. coherent-lemma-i-star-equivalence (cross-chapter, named). O_Delta is coherent on the product.
- Box surjection. lemma-on-product L26469 (carded this batch). Iterated surjections build the resolution.

Source reference. Original reference 087T: focused-seeds/geometry/S9-eqv-diagonal-resolution/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FZ0: sources/stacks-equiv/pinned-original-source.tex, source lines 1750–1750. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let R be a Noetherian ring. Let X be a separated finite type scheme over R which has the resolution property. Set $\mathcal{O}_\Delta = \Delta_*(\mathcal{O}_X)$ where $\Delta : X \to X \times_R X$ is the diagonal of X/k. There exists a resolution*

$$\ldots \to \mathcal{E}_2 \boxtimes \mathcal{G}_2 \to \mathcal{E}_1 \boxtimes \mathcal{G}_1 \to \mathcal{E}_0 \boxtimes \mathcal{G}_0 \to \mathcal{O}_\Delta \to 0$$

where each $\mathcal{E}_i$ and $\mathcal{G}_i$ is a finite locally free $\mathcal{O}_X$-module.

Proof. Since X is separated, the diagonal morphism Δ is a closed immersion and hence $\mathcal{O}_\Delta$ is a coherent $\mathcal{O}_{X \times_R X}$-module (Cohomology of Schemes, Lemma 087T). Thus the lemma follows immediately from Lemma 0FZ0. □

Date: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.